Answers learning task 5

2026 – 2027

Exercise 4.1: Suppose that $f(x) = 2e^{-2x}$ for $0 < x$. Determine the following:

(a) $P(1 < X)$
(b) $P(1 < X < 2.5)$
(c) $P(X = 3)$
(d) $P(X < 4)$
(e) $P(3 \leq X)$
(f) Determine x such that $P(x < X) = 0.20$.
(g) Determine x such that $P(X \leq x) = 0.10$.

Answer

This function is a probability density function of a continuous random variable X (specifically, the exponential distribution). We can compute probabilities on intervals by using integrals and probabilities for specific outcomes are always 0.

(a) $P(1 < X) = \int_{x=1}^{\infty} 2e^{-2x}\,\mathrm{d}x \approx 0.1353$,

(b) $P(1 < X < 2.5) = \int_{x=1}^{2.5} 2e^{-2x}\,\mathrm{d}x \approx 0.1286$,

(c) $P(X = 3) = 0$ (as X is a continuous random variable!),

(d) $P(X < 4) = \int_{x=0}^{4} 2e^{-2x}\,\mathrm{d}x \approx 0.9997$,

(e) $P(3 \leq X) = \int_{x=3}^{\infty} 2e^{-2x}\,\mathrm{d}x \approx 0.00025$,

(f) To determine x such that $P(x < X) = 0.20$ (or $P(X \leq x) = 0.80$), we need to use the "numeric solver":

$$E_1 = \int_0^x 2e^{-2x}\,\mathrm{d}x,$$
$$E_2 = 0.8.$$

The numeric solver gives us $x \approx 0.8047$,

(g) To determine x such that $P(X \leq x) = 0.10$, we need to use the "numeric solver":

$$E_1 = \int_0^x 2e^{-2x}\,\mathrm{d}x,$$
$$E_2 = 0.1.$$

The numeric solver gives us $x \approx 0.0527$.

Exercise 4.7: Suppose that $f(x) = \frac{3}{2}x^2$ for $-1 < x < 1$. Determine the following:

(a) $P(0 < X)$
(b) $P(0.5 < X)$
(c) $P(-0.5 \leq X \leq 0.5)$
(d) $P(X < -2)$
(e) $P(X < 0 \text{ or } X > -0.5)$
(f) x such that $P(x < X) = 0.05$.

Answer

This function is a probability density function of a continuous random variable X. We can compute probabilities on intervals by using integrals and probabilities for specific outcomes are always 0.

(a) $P(0 < X) = \int_{x=-1}^{0} \frac{3}{2}x^2 \, \mathrm{d}x = 0.5$ (notice that the distribution is symmetric around $x = 0$, so we did not need to compute this integral to see this),

(b) $P(0.5 < X) = \int_{x=0.5}^{1} \frac{3}{2}x^2 \, \mathrm{d}x = 0.4375$,

(c) $P(-0.5 \leq X \leq 0.5) = \int_{x=-0.5}^{0.5} \frac{3}{2}x^2 \, \mathrm{d}x = 0.125$,

(d) $P(X < -2) = 0$ (all of the density is between -1 and 1, so this probability is 0),

(e) $P(X < 0 \text{ or } X > -0.5) = 1$ (notice that all possible outcomes are either smaller than zero, or larger than -0.5),

(f) To determine x such that $P(x < X) = 0.05$ (or $P(X \leq x) = 0.95$), we need to use the "numeric solver":

$$E_1 = \int_{-1}^{x} \frac{3}{2}x^2 \, \mathrm{d}x,$$
$$E_2 = 0.95.$$

The numeric solver gives us $x \approx 0.9655$.

Exercise 4.11: The probability density function of the length of a metal rod is $f(x) = 2$ for $2.3 < x < 2.8$ meters.

(a) If the specifications for this process are from 2.25 to 2.75 meters, what proportion of rods fail to meet the specifications?

(b) Assume that the probability density function is $f(x) = 2$ for an interval of length 0.5 meters. Over what value should the density be centered to achieve the greatest proportion of rods within specifications?

Answer

(a) Notice that this is the probability density function of the continuous uniform distribution between $a = 2.3$ and $b = 2.8$. The rods that fail to meet the specifications are either (i) smaller than 2.25 meter or (ii) longer than 2.75 meters. Only the second case is possible. The proportion of rods failing to meet the specifications is

$$P(X \geq 2.75) = \int_{x=2.75}^{2.8} 2 \, \mathrm{d}x = 2 \cdot 0.05 = 0.1.$$

(b) All rods would achieve the specifications if the distribution would be from 2.25 to 2.75 (this is indeed an interval of length 0.5 meters). In other words, the density should be centered at $\frac{2.25+2.75}{2} = 2.5$.

Exercise 4.13: Suppose that the cumulative distribution function of the random variable X

is

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2x & 0 \le x < 5 \\ 1 & 5 \le x \end{cases}$$

Determine the following:

(a) $P(X < 2.8)$
(b) $P(X > 1.5)$
(c) $P(X < -2)$
(d) $P(X > 6)$

Answer

We can use the specified CDF to read the desired probabilities:

(a) $P(X < 2.8) = P(X \le 2.8) = F(2.8) = 0.2 \cdot 2.8 = 0.56$,

(b) $P(X > 1.5) = 1 - P(X \le 1.5) = 1 - F(1.5) = 1 - 0.2 \cdot 1.5 = 0.7$,

(c) $P(X < -2) = F(-2) = 0$,

(d) $P(X > 6) = 1 - P(X \le 6) = 1 - F(6) = 1 - 1 = 0$.

Exercise 4.15: Determine the cumulative distribution function given the following probability density function:

$$f(x) = 2e^{-2x} \qquad \text{for } 0 < x$$

Answer

Notice that $f(x) = 0$ for all $x \le 0$, so is $F(x) = 0$ for all $x \le 0$. For $x > 0$, notice that the cumulative distribution function is the antiderivative of the probability density function:

$$\begin{aligned} F(x) = P(X \le x) = \int_{y=0}^{x} f(y)\,\mathrm{d}y &= \int_{y=0}^{x} 2e^{-2y}\,\mathrm{d}y \\ &= \left[-e^{-2y}\right]_{y=0}^{x} \\ &= -e^{-2x} - (-e^{-2\cdot 0}) \\ &= 1 - e^{-2x}. \end{aligned}$$

Therefore, the cumulative distribution function is equal to

$$F(x) = \begin{cases} 0 & x \le 0 \\ 1 - e^{-2x} & x \ge 0 \end{cases}$$

Exercise 4.21: Determine the cumulative distribution function for the following probability density function:

$$f(x) = \begin{cases} 2, & \text{for } 2.3 < x < 2.8 \\ 0, & \text{otherwise.} \end{cases}$$

Use the cumulative distribution function to determine the probability that a length exceeds 2.7 meters.

Answer

Notice that $f(x) = 0$ for all $x \leq 2.3$, so $F(x) = 0$ for all $x \leq 2.3$. For $2.3 < x < 2.8$, the cumulative distribution function is the antiderivative of the probability density function:

$$\begin{aligned} F(x) = P(X \leq x) = \int_{y=2.3}^{x} f(y)\,\mathrm{d}y &= \int_{y=2.3}^{x} 2\,\mathrm{d}y \\ &= [2y|_{y=2.3}^{x} \\ &= 2 \cdot (x - 2.3). \end{aligned}$$

For $x \geq 2.8$, $f(x) = 0$, so we have reached the peak of $F(x) = 1$ for $x \geq 2.8$.

Therefore, the cumulative distribution function is equal to

$$F(x) = \begin{cases} 0 & x \leq 2.3 \\ 2 \cdot (x - 2.3) & 2.3 \leq x \leq 2.8 \\ 1 & x \geq 2.8 \end{cases}$$

The probability that a length exceeds 2.7 meters is then equal to

$$P(X > 2.7) = 1 - P(X \leq 2.7) = 1 - F(2.7) = 1 - 2 \cdot (2.7 - 2.3) = 0.2.$$

Exercise 4.29: Suppose that $f(x) = \frac{3}{2}x^2$ for $-1 < x < 1$. Determine the mean and variance of X.

Answer

Using the general formulas for expected value and variance of a random variable X, we get:

$$\begin{aligned} E(X) = \int_{x=-\infty}^{\infty} x \cdot f(x)\,\mathrm{d}x &= \int_{x=-1}^{1} x \cdot \frac{3}{2}x^2\,\mathrm{d}x \\ &= \int_{x=-1}^{1} \cdot \frac{3}{2}x^3\,\mathrm{d}x \\ &= \left[\frac{3}{8}x^4\right|_{x=-1}^{1} \\ &= \frac{3}{8} - \frac{3}{8} \\ &= 0. \end{aligned}$$

$$\begin{aligned} V(X) = \int_{x=-\infty}^{\infty} (x - E(X))^2 \cdot f(x)\,\mathrm{d}x &= \int_{x=-1}^{1} (x-0)^2 \cdot \frac{3}{2}x^2\,\mathrm{d}x \\ &= \int_{x=-1}^{1} \cdot \frac{3}{2}x^4\,\mathrm{d}x \\ &= \left[\frac{3}{10}x^5\right|_{x=-1}^{1} \\ &= \frac{3}{10} - (-\frac{3}{10}) \\ &= \frac{6}{10} = 0.6. \end{aligned}$$

Exercise 4.37: Integration by parts is required. The probability density function of the diameter of a drilled hole in millimeters is $f(x) = 10e^{-10(x-5)}$ for $x > 5$ mm. Although the target diameter is 5 millimeters, vibrations, tool wear, and other nuisances produce diameters greater than 5 millimeters.

(a) Determine the mean and variance of the diameter of the holes.

(b) Determine the probability that a diameter exceeds 5.1 millimeters.

Answer

Using the general formulas for expected value and variance of a random variable X, we get:

$$\begin{aligned} E(X) &= \int_{x=-\infty}^{\infty} x \cdot f(x)\, \mathrm{d}x \\ &= \int_{x=5}^{\infty} 10xe^{-10(x-5)}\, \mathrm{d}x. \end{aligned}$$

Using integration by parts, with

$$u = x, \qquad dv = 10e^{-10(x-5)}\, \mathrm{d}x,$$

we obtain

$$du = dx, \qquad v = -e^{-10(x-5)}.$$

Hence,

$$\begin{aligned} E(X) &= \left[-xe^{-10(x-5)}\right]_{x=5}^{\infty} - \int_{x=5}^{\infty} -e^{-10(x-5)}\, \mathrm{d}x \\ &= 5 + \left[-\frac{1}{10}e^{-10(x-5)}\right]_{x=5}^{\infty} \\ &= 5 + \frac{1}{10} = 5.1. \end{aligned}$$

We use the identity

$$V(X) = E(X^2) - (E(X))^2.$$

First, calculate the second moment:

$$E(X^2) = \int_{5}^{\infty} 10x^2e^{-10(x-5)}\, \mathrm{d}x.$$

Applying integration by parts with

$$u = x^2, \qquad dv = 10e^{-10(x-5)}\, \mathrm{d}x,$$

gives

$$du = 2x\, \mathrm{d}x, \qquad v = -e^{-10(x-5)}.$$

Consequently,

$$\begin{aligned} E(X^2) &= \left[-x^2 e^{-10(x-5)}\right]_{x=5}^{\infty} + 2\int_{x=5}^{\infty} x e^{-10(x-5)}\,\mathrm{d}x \\ &= 25 + \frac{2}{10}E(X) \\ &= 25 + \frac{2}{10}(5.1) \\ &= 26.02. \end{aligned}$$

Thus,

$$\begin{aligned} V(X) &= E(X^2) - (E(X))^2 \\ &= 26.02 - (5.1)^2 \\ &= 26.02 - 26.01 = 0.01. \end{aligned}$$

We calculate the probability by integrating the density:

$$\begin{aligned} P(X > 5.1) &= \int_{x=5.1}^{\infty} 10 e^{-10(x-5)}\,\mathrm{d}x \\ &= \left[-e^{-10(x-5)}\right|_{x=5.1}^{\infty} \\ &= e^{-10(5.1-5)} \\ &= e^{-1} \approx 0.3679. \end{aligned}$$

Exercise 4.41: The thickness of a flange on an aircraft component is uniformly distributed between 0.90 and 1.05 millimeters.

(a) Determine the cumulative distribution function of flange thickness.

(b) Determine the proportion of flanges that exceeds 1.02 millimeters.

(c) What thickness is exceeded by $90\,\%$ of the flanges?

(d) Determine the mean and variance of flange thickness.

Answer

The cumulative distribution function of a continuous uniform distribution from a to b is equal to

$$F(x) = \begin{cases} 0, & \text{for } x < a \\ \frac{x-a}{b-a}, & \text{for } a \le x < b \\ 1, & \text{for } x \ge b \end{cases}$$

(a) With other words, the cumulative distribution function for $a = 0.90$ and $b = 1.05$ equals

$$F(x) = \begin{cases} 0, & \text{for } x < 0.90 \\ \frac{x-0.90}{0.15}, & \text{for } 0.90 \le x < 1.05 \\ 1, & \text{for } x \ge 1.05 \end{cases}$$

(b) $P(X > 1.02) = 1 - P(X \le 1.02) = 1 - F(1.02) = 1 - \frac{1.02-0.90}{0.15} = 0.2$,

(c) The thickness exceeded by $90\,\%$ of the flanges is at a tenth of the interval from 0.90 to 1.05. With other words, this thickness is equal to $a + 0.1(b - a) = 0.90 + 0.1(1.05 - 0.9) = 0.915$.

(d) The mean and variance of a uniform distribution are equal to $\frac{a+b}{2}$ and $\frac{(b-a)^2}{12}$ respectively. Therefore, we have:

$$E(X) = \frac{0.9 + 1.05}{2} = 0.975,$$
$$V(X) = \frac{(1.05 - 0.9)^2}{12} = 0.001875.$$